ABHISHEK KUMAR LAL

PHONE
+91 8691066937

EMAIL
LALABHISHEKKUMAR241@GMAIL.COM

ADDRESS
MUMBAI, MAHARASHTRA

ABOUT ME

Versatile and driven Engineer with hands-on experience in both embedded systems and software development. Skilled in building real-time robotics and IoT systems using Arduino, ESP32, and Raspberry Pi, as well as developing Python-based automation tools, APIs, and web applications. Proven ability to bridge the gap between hardware and software through innovative project execution, clean code, and system integration. Adept at working in agile teams, with strong debugging, prototyping, and problem-solving skills across domains.

CERTIFICATION

- Arduino Workshop – 2023
- Robo Workshop – 2023
- CMOS & FinFET Cell Design – 2023
- TaraNG Robotics Certification – 2024

SKILL

- **Languages:** C, C++, Python, HTML, CSS, SQL
- **Microcontrollers:** Arduino, NodeMCU, ESP32
- **Platforms:** Raspberry Pi (Linux, Python), Blynk, ThingSpeak, AWS
- **Communication:** UART, I2C, SPI, Wi-Fi, Bluetooth
- **Tools:** Arduino IDE, KiCad (Basics), Multisim, VS Code
- **Cloud:** Blynk, Firebase (IoT Control)

EXTRAS

- Organizer of tech workshops on Wi-Fi controlled robots and NodeMCU IoT.
- College committee core member and robotics competition host.

EDUCATION

BE Electronics and Telecommunication
A. C. Patil College Of Engineering (ACPCE)
Kharghar, Navi-Mumbai, Maharashtra — *2025/CGPA - 8*

HSC in Science
Tilaknagar Junior College
Dombivli, Maharashtra — 2021/71%

SSC
R B T Vidyalaya
Dombivli, Maharashtra — 2019/64.35%

EXPERIENCE

Embedded Engineer
PNT Robotics and Automation Solutions – Dombivli, Maharashtra
March 2025 – Present

- Working on intelligent robotic systems with firmware integration and sensor interfacing.
- Implementing software logic for embedded devices with optimized memory handling.
- Implemented sensor-based logic and communication protocols on microcontrollers.

Trainee Engineer
Inbox Creations Pvt Ltd – Kurla, Mumbai, maharashtra
June 2023 – Dec 2023

- Developed backend logic for automation projects using Arduino and Python.
- Delivered code and system diagrams as per client requirements.
- Involved in project documentation and version control using Git.

PROJECTS

- **Wi-Fi Controlled Car using ESP32 -** Built a mobile robot controlled wirelessly via mobile app using ESP32 and motor drivers.
- **Robotic Arm for Advertisement -** Created a precision-controlled robotic arm for product showcasing in industrial setups.
- **Vending Machine using Arduino -** Developed automated product dispensing machine for retail settings.
- **Alcohol Detection System with MQ-3 -** Safety solution that detects alcohol levels and could be integrated with vehicle ignition systems.